Software Configuration Management – 3

Lecture # 35

Recap

Today's Lecture

- We'll discuss some practical aspects related to software configuration management in today's lecture

- It was mentioned in the first lecture on software configuration management that SCM provides a cover against lack of visibility and lack of traceability
- So, now we'll discuss how SCM infuses visibility and traceability through its main functions

SCM Functions Infuse Visibility

Visibility: Identification

- User/buyer/seller can see what is being/has been built/is to be modified
- Management can see what is embodied in a product
- All project participants can communicate with a common frame of reference

Visibility: Control

- Current and planned configuration generally known
- Management can see impact of change
- Management has option of getting involved with technical detail of project

Visibility: Auditing

- Inconsistencies and discrepancies manifest
- State of product known to management and product developers
- Potential problems identified early

Visibility: Accounting/Reporting

- Reports inform as to status
- Actions/decisions made explicit (e.g., through CCB meeting minutes)
- Database of events is project history

SCM Functions Infuse Traceability

Traceability: Identification

- Provides pointers to software parts in software products for use in referencing
- Make software parts and their relationships more visible, thus facilitating the linking of parts in different representations of the same product

Traceability: Control

- Makes baselines and changes to them manifest, thus providing the links in a traceability chain
- Provides the forum for avoiding unwanted excursions and maintaining convergence with requirements

SCM Infuses Traceability

Traceability: Auditing

- Checks that parts in one software product are carried through to the subsequent software product
- Checks that parts in a software product have antecedents/roots in requirements documentation

Traceability: Accounting/Reporting

- Provides history of what happened and when
- Provides explicit linkages between change control forms

Real-World Considerations

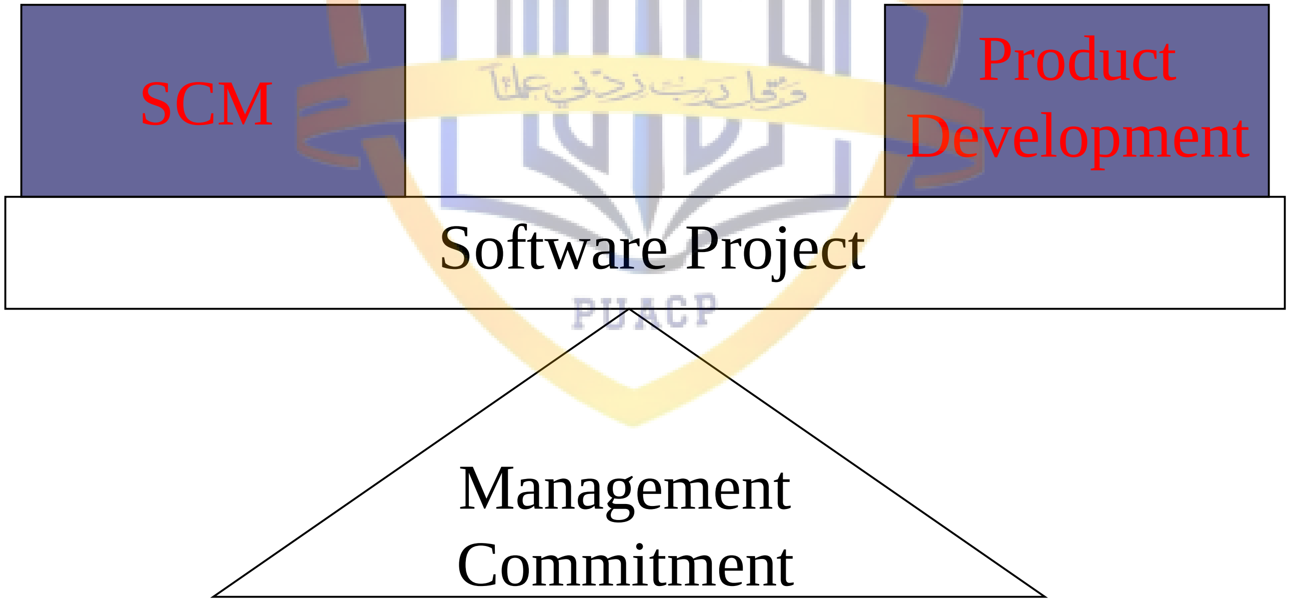
Real-World Considerations

- Management Commitment
- SCM Staffing
- Establishment of a CCB
- SCM During the Acceptance Testing Cycle
- Justification and Practicality of Auditing
- Avoiding the Paperwork Nightmare
- Allocating Resources among SCM Activities

Management Commitment

- Management commitment to the establishment of checks and balances is essential to achieving benefits from SCM
- (say many things here)

Management Commitment for SCM



SCM Staffing - 1

- Initial staffing by a few experienced people quickly gains the confidence and respect of other project team members
- It is important to build the image that the SCM team has the objective of helping the other project team members achieve the overall team goals, that they are not a group of obstructionists and criticizers

SCM Staffing - 2

- An SCM organization requires
  - Auditors
  - Configuration control specialists
  - Status accountants
- These positions require hard work and dedication

SCM Staffing - 3

- Important qualification of these people are the ability to see congruence (similarity) between software products and the ability to perceive what is missing from a software product
- With these abilities, the SCM team member can observe how change to the software system is visibly and traceably being controlled

SCM Staffing - 4

- Should all SCM personnel be skilled programmers and analysts?
- No, these particular skills are not a necessity by any means, although personnel performing software configuration auditing should be technically skilled

SCM Staffing Skills: Identification

- Ability to see partitions
- Ability to see relationships
- Some technical ability desirable
  - System engineering orientation
  - Programming

SCM Staffing Skills: Control

- Ability to evaluate benefits versus costs
- System viewpoint (balance of technical / managerial, user / buyer / seller)
- An appreciation of what is involved in engineering a software change

SCM Staffing Skills: Auditing

- Extreme attention to detail
- Ability to see congruence
- Ability to perceive what is missing
- Extensive experience with technical aspects of system engineering and/or software engineering

SCM Staffing Skills: Status Accounting/Reporting

- Ability to take notes and record data
- Ability to organize data
- Some technical familiarity desirable but not required
  - System engineering orientation
  - Programming

Establishment of a CCB - 1

- As a starting point in instituting SCM, periodic CCB meetings provide change control, visibility, and traceability
- The CCB meeting is a mechanism for controlling change during the development and maintenance of software
- CCB membership should be drawn from all organizations on the project

Establishment of a CCB - 2

- Decision mechanism
- CCB chairperson
- CCB minutes

- When changes are necessary, a CCB meets to evaluate and manage the impact of change on the software development process

- The impact of change can be countered in only three ways
  - Add more people to the project to reduce the impact of the change
  - Extend the time to completion
  - Eliminate other nonessential or less essential functionality

- If a small amount of code is changed, it is redesigned into the old code; if a large amount is changed, a complete subsystem is redesigned as though it were a new product
- From a maintenance point of view, IBM followed this rule of thumb; “If 20% of the code must be modified, then the module should be redesigned and rewritten”

SCM During the Acceptance Testing Cycle

- SCM integrated within the acceptance testing cycle maintains a visible and traceable product ready for delivery to the customer

Testing: Identification

- Preparation of release notes (lists of changed software)
- Identification of development baseline
- Identification of incident reports
- Identification of operational baseline

Testing: Control

- CCB meetings
  - Establishment of development baseline
  - Assignment of testing and incident resolution priorities
  - Establishment of turnover dates
  - Approval of audit and test reports
  - Approval of incident report resolutions
  - Establishment of operational baseline

Testing: Auditing

- Comparison of new baseline to previous baseline
- Assurance that standards have been met
- Testing (verification and validation) of software system

Testing: Accounting/Reporting

- Logging and tracking of incident reports
- Publication of CCB minutes

Justification and Practicality of Auditing

- Although the auditing consumes the greater part of the SCM budget, it has the potential of preventing the waste of much greater resources

Avoiding the Paperwork Nightmare

- The buyer/user and seller should agree on the paperwork needed to achieve a mutually desirable level of visibility and traceability

Allocating Resources among SCM Activities

- Cost versus benefits must be evaluated for each individual project in determining the allocation of limited SCM resources

Important Issues Relating to SCM

- Following issues should be examined and evaluated in terms of corporate return on investment and employee leverage
  - How long to support a particular version of the software?
  - What upgrade paths should be allowed?
  - How many variations (not versions) of the product should be produced and supported?

References

- Hand Book of Software Quality Assurance 3rd Ed., Edited by G. Gordon Schulmeyer and James I. McManus, 1998 (Chapter 10.2-10.4)